# Ahlam Zahran

AI Engineer & Data Scientist
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Amman, Jordan | U.S. citizen | Open to relocation (Saudi Arabia, UAE, U.S.)

## Summary

AI engineer and data scientist with hands-on experience training, evaluating and deploying computer-vision and generative models. Rebuilt a CVPR 2023 fMRI-to-image pipeline reaching **82.1%** identification accuracy (chance 50%, $p < 0.001$) and shipped it as a full-stack web application. Strong in statistical rigour and end-to-end delivery.

## Education

**University of Petra** — Amman, Jordan
*B.Sc. in Artificial Intelligence & Data Science | GPA: 3.59/4.00* — *Feb 2022 – Jul 2026*

**German Jordanian University** — Amman, Jordan
*Biomedical Engineering (transferred)* — *Sep 2019 – Jun 2022*

## Experience

**Neurender (research project)** — Amman, Jordan
*Founder & Lead Researcher* — *2026 – Present*

- Leading independent research continuing NeuroSeekers: AI that reconstructs the image a person is seeing from fMRI brain scans, with every result scored, bounded and traceable.
- Extending decoding to new people without retraining (cross-person decoding), working toward validation on standard 3-Tesla hospital scanners and a public verification standard for brain-decoding claims.

**NeuroSeekers** — University of Petra
*AI Engineer & Full-Stack Developer (graduation project)* — *2025 – 2026*

- Rebuilt the Takagi et al. (CVPR 2023) fMRI-to-image pipeline with text-guided semantic decoding, reaching **82.1%** two-way identification accuracy on 982 held-out images (chance 50%; $p < 0.001$ vs. 1,000 shuffled controls).
- Trained subject-specific decoders on the Natural Scenes Dataset (7T fMRI, 26,688 voxels per image): ROI-wise encoding, ridge regression and latent-space decoding; Stable Diffusion image generation with a blind candidate selector.
- Built the evaluation framework: measured noise ceiling, 16 pre-registered variants and a provenance record for every result.
- Shipped a full-stack web app (Django, React) with a live demo and real-time analysis, ∼30 seconds per image.
- Presented at **SOFEX Jordan 2025** (under the patronage of HRH Crown Prince Al Hussein bin Abdullah II) and **SOFEX Aqaba 2026**; covered by Jordan TV, Roya TV, AlMamlaka TV, Amen FM and The Jordan Times.

**Bait Misk | The Concept Coffee Store** — Jordan
*Supply Chain Manager* — *Apr 2026 – Jun 2026*

- Built a cost-management system from scratch (13+ suppliers, 56+ ingredient unit costs, per-item menu costing) and tracked thousands of SKUs with Excel and Power BI dashboards.

**IEEE Computational Intelligence Society** — University of Petra
*Student Chapter Chair* — *2024*

- Led technical events, workshops and outreach; also taught hands-on Arduino IoT workshops (2024).

## Projects & Awards

**Autonomous Humanitarian Demining Robot** — 2026
*2nd Place, World Robot Caspian Cup (WRCC), Open Category, Baku*

- Autonomous robot combining computer vision, LiDAR, GPS, ROS2 and sensor fusion for safer mine detection.

**Peer-to-Peer Energy Trading Platform** — 2024
*1st Place, WEFE Hackathon*

- Reinforcement learning and a greedy algorithm to optimize renewable-energy distribution; full UI/UX prototype in Figma.

**AWS DeepRacer** — 2023
*Top 500 globally*

- Designed and trained autonomous racing models with Proximal Policy Optimization (PPO) on AWS SageMaker.

## Technical Skills

**Languages**: Python, SQL, Java, JavaScript
**AI & ML**: PyTorch, TensorFlow, scikit-learn, computer vision, diffusion models (Stable Diffusion), NLP, reinforcement learning, RAG
**Data & Neuroimaging**: fMRI analysis (NSD), ridge-regression decoding, statistical testing, Power BI, data warehousing
**Deployment & Cloud**: Django, React, REST APIs, Docker, AWS SageMaker, AWS S3, Git/GitHub
**Robotics & IoT**: ROS2, Arduino, ESP32, FPGA/VHDL

## Certifications & Languages

**Certifications:** Generative AI with AWS, Udacity (2024); DAAD Business Training, Germany (2024); Erasmus+ RL4Eng, Bonn-Rhein-Sieg University (2023); Multi-Robot Drone Workshop, Sager (2023).
**Languages:** Arabic (native), English (fluent), German (intermediate)